Numeric Literals

Representing numbers in Python

Integer Literals (int)

Whole numbers, both positive and negative, without any decimal points. Used for countable quantities.

```
# Number of floors in a skyscraper
number_of_floors = 55

# Basement levels (negative integer)
underground_levels = -4

# Total rooms
total_rooms = 1200
```

Float Literals (float)

Real numbers that contain a decimal point. Essential for measurements requiring precision.

```
# Height of a door in meters
door_height = 2.1

# Site area in hectares
site_area = 1.5

# Concrete strength in MPa
concrete_mpa = 25.5
```

Complex Literals (complex)

Numbers with a real and imaginary part (ending in 'j'). Used in specialized engineering fields like electrical or vibration analysis.

```
# An electrical impedance value
impedance = 12.5 + 3.2j

# A value from a vibration analysis
damping_factor = 0.95 + 0.1j
```